Trinkey Sampling, BNO055 Timing, and Compensator Evaluation

1 Recommended decision

Use a nominal paired-sensor acquisition rate of 250 Hz with the BNO055 accelerometer bandwidth set to 125 Hz for the planned 10 Hz and 35 Hz print comparisons. Retain Klipper's 1 kHz controller grid. These rates form a valid multirate system: the command may be generated at 1 kHz while measured acceleration arrives every 4 ms.

For the present evaluation, do not invent the three intervening acceleration values between sensor samples. Compare the desired acceleration with the measured acceleration at the genuine, time-mapped BNO055 sample times. This gives a valid *acceleration-domain* tracking metric. It does not reproduce the paper's position-domain RMS tracking error unless an observer or another position estimate is added.

The two principal evaluation outputs should therefore be:

1. printed-part evidence: photographs and repeated dimensional measurements; and
2. a time-aligned acceleration-domain comparison from Motan, reported as an adaptation necessitated by the available instrumentation.

This is a defensible dissertation limitation. It avoids claiming that interpolated data contain information that the sensors did not measure.

2 What the current acquisition actually measures

The streaming record contains separate timestamps for the base/bed and toolhead accelerometers. The firmware reads the base sensor and then the toolhead sensor over I^2C. For each sensor, the RP2040 timestamp is recorded *after* the register read succeeds. The host then maps that timestamp from the Trinkey clock to Klipper print time using the fitted affine clock relation

$$t_{\text{print}} = \alpha t_{\text{Trinkey}} + \beta. \tag{1}$$

This design is better than assigning one shared timestamp to both sensors, and it removes host USB arrival time from the signal alignment. However, the timestamp is not the instant at which the BNO055 internally sampled the MEMS element or updated its output register. The BNO055 does not provide a per-sample timestamp register. Consequently, the remaining timing uncertainty includes the age of the register value, the sequential read time, and the sensor's internal filter delay. Exact data-ready timing would require each BNO055 interrupt/data-ready output to be wired to a separate RP2040 GPIO and timestamped at the edge. Even then, the edge would identify data availability, not the exact physical centre of the filtered measurement [2].

Given the remaining project time, retaining the present hardware and reporting this uncertainty is preferable to changing the accelerometers. The same sensor and timing path is used in every controller condition, so repeatable latency is largely common to the comparisons, although it must still be treated consistently when computing phase-sensitive errors.

3 Why the requested 400 Hz stream was not 400 Hz

The reported status at the end of the supplied run contained 45,438 scheduled sample indices and 18,266 missed indices. Thus only

$$45{,}438 - 18{,}266 = 27{,}172 \tag{2}$$

paired acquisitions were performed, or approximately 60 % of the requested samples. The approximately 3.8 ms worst acquisition loop cannot meet the 2.5 ms period required by 400 Hz. The absence of transport, host-queue, packet, and frame errors shows that the bottleneck was the sequential sensor acquisition rather than USB transfer or Motan.

At 250 Hz, the available period becomes 4 ms, which is only about 0.2 ms above the observed worst loop time. Therefore 250 Hz is a candidate operating point, not an assumption. Before accepting a run, verify that the following counters do not increase throughout a complete print:

$$\texttt{firmware_dropped} = \texttt{firmware_overruns} = \texttt{firmware_transport_dropped} = 0, \tag{3}$$

and also require zero host queue drops, packet-sequence errors, frame errors, and read errors. A batch size of four concerns transport only; it does not mean that four samples share one timestamp.

The 250 Hz stream provides 25 samples per cycle at 10 Hz and about 7.1 samples per cycle at 35 Hz. This is adequate for the intended fundamental-frequency comparison. It would provide only 2.5 samples per cycle at 100 Hz; Nyquist is satisfied, but the margin is weak and broadband results near 100 Hz should not be presented as high-fidelity measurements. The 125 Hz BNO055 bandwidth setting also leaves limited anti-alias margin at a 250 Hz acquisition rate. For the present tests, frequency-selective results at 10 Hz and 35 Hz are more defensible than a full-band RMS claim extending to 100 Hz.

4 Evaluation procedure

The generated trajectories provide matched uncompensated and filtered B-spline prints at 10 Hz and 35 Hz. Their motion and extrusion payloads are identical at a given frequency; only the controller state changes. This supports a controlled two-by-two comparison.

4.1 Dynamic metric from Motan

The Motan toolhead TrapQ is the nominal desired trajectory. The filtered B-spline wrapper modifies the stepper kinematic position without replacing that nominal TrapQ. Consequently, the reference should be `trapq(toolhead, x_accel)` or `trapq(toolhead, y_accel)`, and the measured output should be:

- toolhead accelerometer X for X-axis motion; and
- bed/base accelerometer Y for Y-axis motion.

For each genuine accelerometer timestamp t_k, evaluate the corresponding TrapQ acceleration directly at t_k. The lightweight capture endpoint does this inside Klipper and stores the result beside the physical sample as `trinkey_accel(toolhead, command_x)` or `trinkey_accel(base, command_y)`. It therefore avoids exporting the complete high-density TrapQ. This is preferable to first creating two independently rounded 250 Hz sequences. If a uniform grid is needed by an analysis tool, resample both signals once onto the same 250 Hz grid after clock mapping and document the interpolation and filtering used.

After applying the measured sensor orientation/sign, scale and static-bias correction, define

$$e_a[k] = a_{\mathrm{ref}}(t_k) - a_{\mathrm{meas}}(t_k), \qquad \mathrm{RMSE}_a = \sqrt{\frac{1}{N}\sum_{k=1}^{N} e_a[k]^2}. \tag{4}$$

Report Equation (4) in $\mathrm{mm\,s^{-2}}$, optionally normalised by the RMS reference acceleration. Also fit the 10 Hz or 35 Hz sine and cosine components to report error amplitude and phase.

This narrow-band result is less sensitive to out-of-band noise and aliasing than an unqualified broadband RMS value.

Use the same steady central-wall interval in every run and exclude warm-up, parking and the real end-cap corner transients. State explicitly whether a constant instrumentation delay was corrected. If no delay correction is made, call the result *end-to-end acceleration error*; this remains useful for relative compensated/uncompensated comparison because the acquisition path is common, but delay contributes to the numerical error.

Do not label Equation (4) as the paper's position tracking error. For a nearly single-frequency error only, an equivalent displacement phasor can be estimated from

$$X_e(j\omega) = -\frac{A_e(j\omega)}{\omega^2}, \qquad \omega = 2\pi f. \tag{5}$$

This conversion must be described as narrow-band; double-integrating a general accelerometer record will otherwise amplify bias and drift. A genuine position-domain RMS comparison requires the Luenberger observer used in the reference study, an equivalently validated estimator, or an independent position measurement.

4.2 Printed-part metric

The proposed physical evaluation follows the reference paper's two forms of evidence: visible print quality and a repeated geometric measurement. For each frequency and controller condition:

1. keep filament, temperature, layer height, flow, acceleration limits, thermal state and all other settings fixed; keep input shaping disabled;
2. photograph the same central wall region with fixed lighting, orientation, distance and scale;
3. measure the nominal 3.4 mm outside width at ten predefined, equally spaced locations in the central region; and
4. report the mean width, standard deviation, mean signed dimensional error, and the measurement locations. Repeat whole prints where time permits so print-to-print variability is not confused with controller performance.

The paper compared uncompensated, standard FBF and hybrid FBF cases at low and high excitation frequencies, using both part measurements and RMS tracking error. The present uncompensated-versus-filtered-B-spline comparison therefore follows the same evaluation structure, but with fewer controller conditions and an acceleration-domain dynamic metric until a position observer is available [1].

5 Implication for the later hybrid controller

A 1 kHz, 140-sample controller window spans 140 ms. A 250 Hz sensor supplies only 35 independent measurement epochs in that window. This is not a problem for the evaluation above: its error sum simply uses the 35 real epochs. It *is* an implementation issue for the hybrid controller if its lifted model expects 140 output values.

The appropriate future solution is a multirate observer: predict the estimated state every 1 ms from the known command and apply a measurement correction whenever a new 250 Hz accelerometer sample arrives. The observer then supplies a 1 kHz estimated output sequence to the lifted controller while preserving the fact that only every fourth step contains new measurement information. Simple interpolation or sample repetition may produce a vector of length 140, but does not create 140 independent observations and should not be presented as such. Timestamped measurement delay must also be included in the observer or batch-history alignment.

Thus the final decision is:

- **baseline compensator evaluation now:** use the real 250 Hz samples, without missing-sample estimation;
- **paper-comparable position error:** add a validated observer; and

- **future hybrid feedback implementation:** retain the 1 kHz controller and use an explicit multirate observer/residual construction rather than treating interpolated samples as measurements.

References

[1] C.-H. Chou, M. Duan, and C. E. Okwudire, "A Physics-Guided Data-Driven Feedforward Tracking Controller for Systems with Unmodeled Dynamics—Applied to 3D Printing," 2022. doi: 10.48550/arXiv.2206.11960.

[2] Bosch Sensortec, "BNO055 Intelligent 9-axis absolute orientation sensor: Data sheet," BST-BNO055-DS000. https://www.bosch-sensortec.com/media/boschsensortec/downloads/datasheets/bst-bno055-ds000.pdf.